EXPERIMENT 2 – STUDY OF NETWORK CABLES

AIM

To study different types of network cables and to make a **straight-through cable and cross-over cable** using an RJ45 connector and crimping tool.

REQUIREMENTS

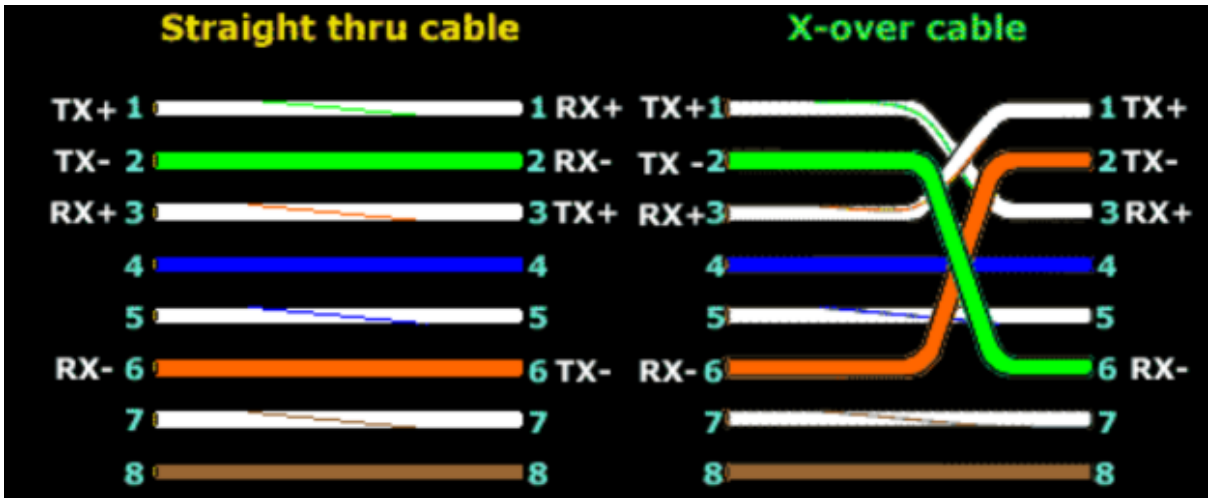
- Ethernet cable (CAT5/CAT5e)
- RJ45 connectors
- Crimping tool
- Cable tester (if available)

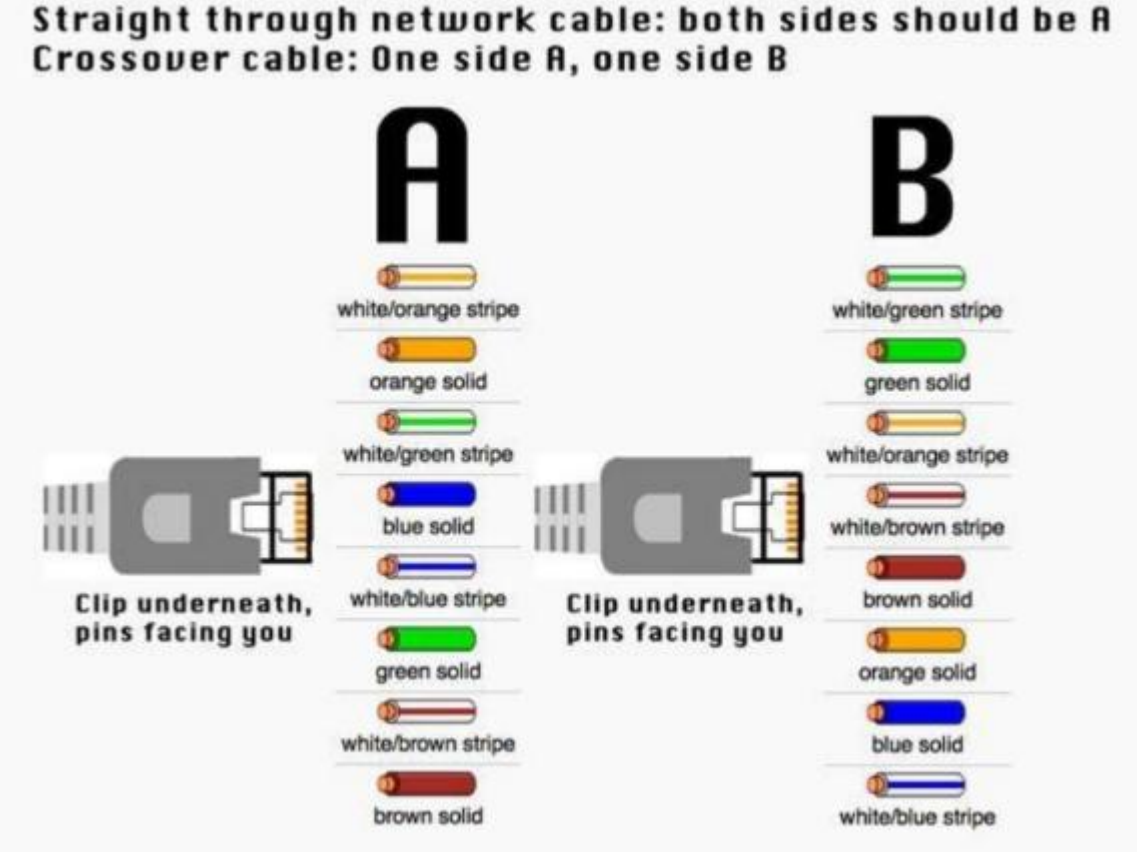
TYPES OF NETWORK CABLES

1. UTP – Unshielded Twisted Pair
2. STP – Shielded Twisted Pair
3. Coaxial Cable
4. Fibre Optic Cable

PROCEDURE

1. Take an Ethernet cable and cut it to the required length.
2. Strip approximately **1.5 cm** of the outer cable covering.
3. Untwist and separate the four pairs of wires.
4. Arrange the wires according to the required wiring standard.
5. For a **straight-through cable**, use the same arrangement at both ends.
6. For a **cross-over cable**, use arrangement A at one end and arrangement B at the other end.
7. Cut the wires evenly and insert them into the RJ45 connector.
8. Ensure that all wires reach the end of the connector in the correct order.
9. Insert the RJ45 connector into the crimping tool and press firmly.
10. Repeat the process for the other end.
11. Test the completed cable by connecting it to the appropriate devices.





OBSERVATION

Cable	Wiring Arrangement	Usage
Straight-through cable	Same standard at both ends	PC to Switch/Router

Cable	Wiring Arrangement	Usage
Cross-over cable	Different standards at each end	PC to PC

RESULT

Thus, the different types of network cables were studied, and **straight-through and cross-over Ethernet cables were successfully prepared using RJ45 connectors and a crimping tool**. The cables were tested and found to be working properly.